Object-Oriented System Analysis Design

2nd September 2024

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

a) Define the following terms:

i) Cohesion **1**

ii) An Actor in the context of use-case modelling. **1**

iii) Cardinality or multiplicity **1**

c) Multiple choice questions: Please select ONE (1) from the options provided

i) Which of the following is NOT a UML diagram used in a system analysis model? **1**

A. use-case diagram
B. class diagram
C. dataflow diagram
D. state diagram

ii) Which of the following would NOT be depicted as an event in a UML state diagram? **1**

A. an input from a user
B. a time-out
C. a message from an object
D. a class

iii) Which of the following items appears on a CRC card? **1**

A. class collaborators
B. class reliability
C. class responsibilities
D. Both A and B
E. Both A and C
F. Both B and C

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Marks

iv) Object Constraint Language (OCL) can be used for **1**

A. Specifying stereotypes and types in the class model
B. Describing pre- and post-conditions on Operations and Methods
C. Describing Guards
D. All the above
E. None of the above

v) In object-oriented design, methods and attributes from a super class being available to a sub-class is called **1**

A. Cohesion
B. Inheritance
C. Decoupling
D. Prioritisation

vi) In OO, a procedure or operation performed by a class is a **1**

A. Sub-class
B. Instantiation
C. Method
D. Attribute

vii) In Agile software processes, once requirements have been agreed, they **1**

A. cannot be changed
B. can be changed
C. can be changed once a running model of the system has been produced
D. can be changed once only

Total 10 Marks

Question 2

a) Discuss the considerations for choosing between Object-Oriented System Analysis and Design (OOSAAD) and Structured Systems Analysis and Design (SSADM). **2**

b) Outline FOUR (4) differences between the OOSAAD and SSADM. **8**

Total 10 Marks

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Marks

Question 3

a) Explain TWO (2) advantages and TWO (2) disadvantages of prototyping. **4**

b) Compare and contrast any TWO (2) of the prototyping methods below. **6**

i. Storyboards
ii. Wizard of OZ
iii. High level programming simulation.

Total 10 Marks

Question 4

a) Differentiate between Blackbox and Whitebox testing. **2**

b) Describe TWO (2) aspects in which OO software testing and traditional non-object-oriented differ and give TWO (2) examples of OO Blackbox or Whitebox testing to help explain the difference. **8**

Total 10 Marks

Question 5

a) Describe modular design and component-based design. **4**

b) Explain at least SIX (6) benefits of these approaches. **6**

Total 10 Marks

Question 6

a) Explain what is meant by a requirements catalogue in the context of system analysis and design. **2**

b) Describe typical contents of a requirements catalogue. **8**

Total 10 Marks

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Marks

Question 7

Draw a class diagram showing FOUR (4) only attributes of a car and SIX (6) only possible whole-part components. **10**

Total 10 Marks

Question 8

Explain the FIVE (5) SOLID principles applicable to Object-Oriented (OO) design. **10**

Total 10 Marks

Question 9

Compare and contrast the following object-oriented analysis and design terms. You must provide examples to illustrate your answers:

a) A Message and a Method **5**

b) Abstract Class and Instance of a class **5**

Total 10 Marks

Question 10

Draw a UML State Diagram to illustrate the following process for an online purchase and delivery of products. **10**

1. Customers select one or more products from a product selection page.
2. Customers select checkout and enter payment details.
    a. If payment is not valid, customers are returned to the Checkout page
    b. If payment is valid, customers select a delivery method (collection or shipping) and complete appropriate details.
3. The system confirms the order and warehouse fulfils the order.

Your response must also demonstrate use of guards, nesting and self-messaging.

Total 10 Marks

End of paper

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